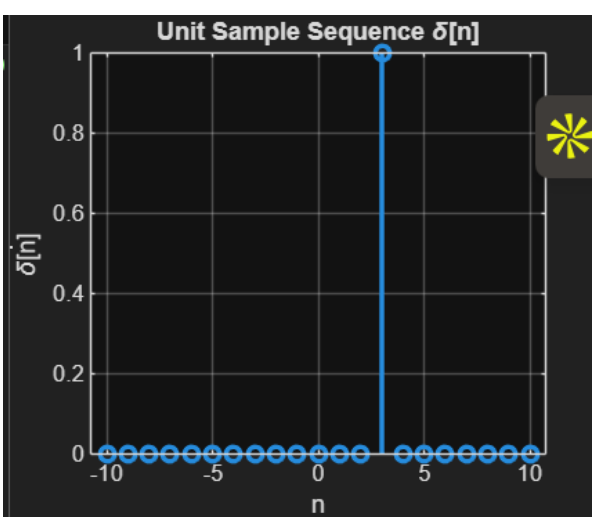
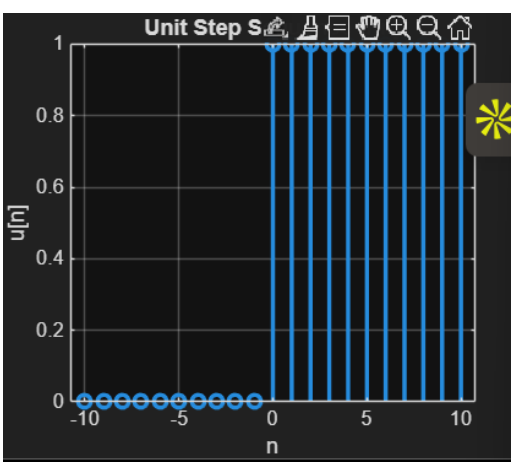
Unit Sample (DFT)

```
% Unit sample
clc;
clear;
close all;
n = -10:10;          % range of n
x = (n == 3);        % unit sample sequence δ[n]
stem(n, x, 'LineWidth', 2);
grid on;
xlabel('n');
ylabel('\delta[n]');
title('Unit Sample Sequence \delta[n]');
```



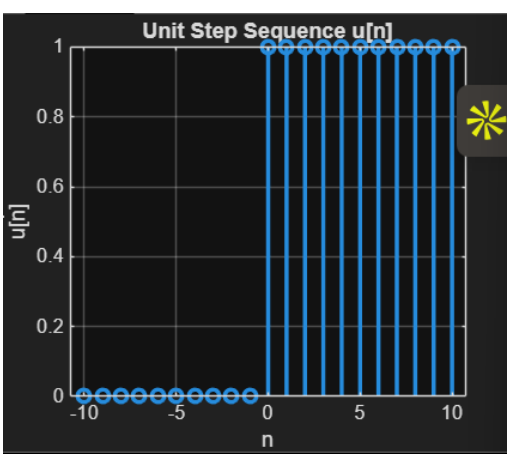
Unit Step (DFT)

```
clc;
clear;
close all;

n = -10:10;          % range of n
x = (n >= 0);        % unit step sequence u[n]

stem(n, x, 'LineWidth', 2);
grid on;
xlabel('n');
ylabel('u[n]');
title('Unit Step Sequence u[n]');
```



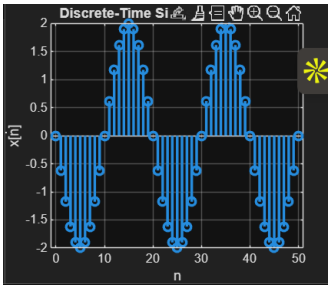
Unit ramp

```
clc;
clear;
close all;

n = -10:10;             % range of n
x = n .* (n >= 0);      % unit ramp sequence r[n]

stem(n, x, 'LineWidth', 2);
grid on;
xlabel('n');
ylabel('r[n]');
title('Unit Ramp Sequence r[n]');
```



Sinosoidal

```
clc;
clear;
close all;
A = 2;              % Amplitude
f = 5;              % Frequency (Hz)
T = 0.01;           % Sampling time (seconds)
phi = pi;           % Phase (radians)
n = 0:50;           % range of n
x = A*sin(2*pi*f*n*T + phi);   % sinusoidal sequence
stem(n, x, 'LineWidth', 2);
grid on;
xlabel('n');
ylabel('x[n]');
title('Discrete-Time Sinusoidal Sequence');
```



DTFT (a(n) . u(n))

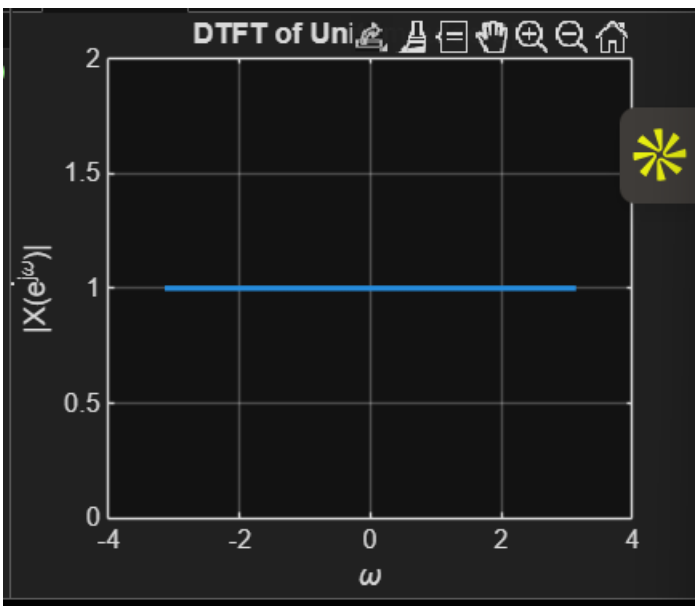
```
clc;
clear;
syms n w a
x = a^n * heaviside(n);
X = symsum(x * exp(-1j*w*n), n, 0, inf);
X = simplify(X);
disp('DTFT of a^n u[n]:');
disp(X);
```

```
DTFT of a^n u[n]:
piecewise(a*exp(-w*1i) == 1, Inf + 1/#E^(w*1i)*a, a*exp(-w*1i) ~= 1, 1/#E^(w*1i)*a + (exp(-
```

Unigt Impulse DTFT

```
clc;
clear;
close all;

w = -pi:0.001:pi;     % frequency range
X = ones(size(w));   % DTFT of delta[n] is 1
```

```
plot(w, abs(X), 'LineWidth', 2);
grid on;
xlabel('\omega');
ylabel('|X(e^{j\omega})|');
title('DTFT of Unit Impulse \delta[n]');
```



Unit Step DTFT

```
clc;
clear;
```

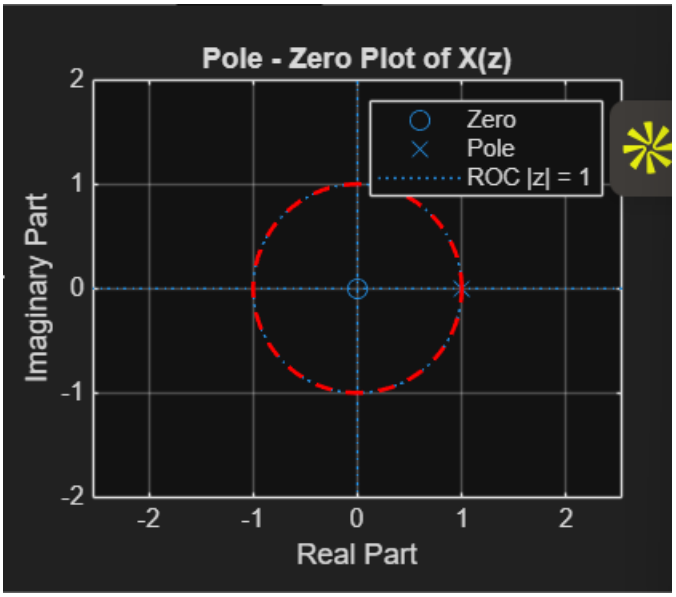
```
syms n w z

u = heaviside(n);

X_z = ztrans(u, n, z);          % Z-transform of unit step
X = subs(X_z, z, exp(1j*w));    % Substitute z = e^(jw) for DTFT
X = simplify(X);

disp('DTFT of Unit Step:');
disp(X);
```

```
DTFT of Unit Step:
1/(exp(w*1i) - 1) + 1/2
```

Z Transform (x(n).u(n))

```
syms n z
X = symsum(z^(-n), n, 0, inf); % Z-transform
disp('Z-transform X(z) = ');
simplify(X)
b = [1 0]; % Coefficients
```

```
a = [1 -1];
% Plot
figure;
zplane(b, a);
title('Pole - Zero Plot of X(z)');
grid on;
hold on;
% ROC (unit circle)
theta = linspace(0, 2*pi, 200);
r = 1;
plot(r*cos(theta), r*sin(theta), 'r--', 'LineWidth', 1.5);
legend('Zero', 'Pole', 'ROC |z| = 1');
xlim([-2 2]);
ylim([-2 2]);
axis equal;
```




```
>> ztransformxnun
Z-transform X(z) =

ans =

piecewise(1 < abs(z), z/(z - 1), abs(z) in Dom::Interval(0, 1) & z in Dom::Interval(0, [1]))
```

ZERO Plot of function

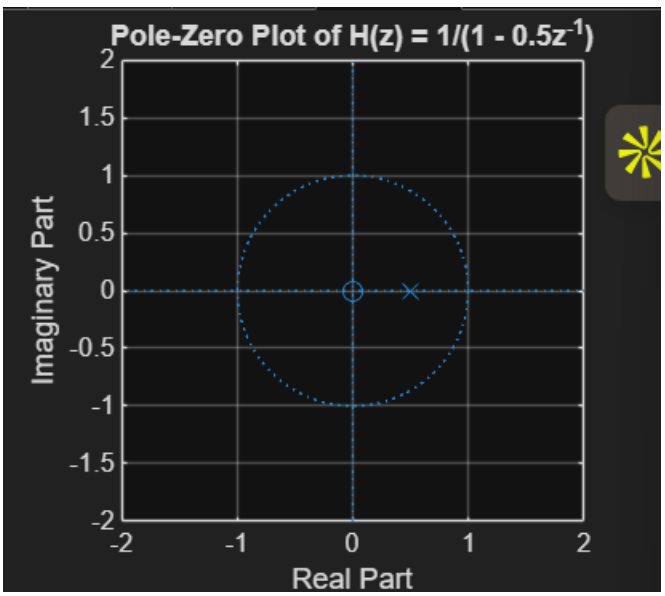
```
% H(z) = 1/(1 - 0.5 z^-1)
% => H(z) = z/(z - 0.5)
b = [1 0];        % numerator: z
a = [1 -0.5];     % denominator: z - 0.5
figure;
zplane(b, a);
grid on;
title('Pole-Zero Plot of H(z) = 1/(1 - 0.5z^{-1})');
```

% Zoom for better view

xlim([-2 2]);

ylim([-2 2]);

% Pole is where the value of the transform gives you infinity



DTFT of function ([0 1 1 1])

clc;

clear;

close all;

```
x = [0 1 1 1];      % input sequence
N = length(x);

n = 0:N-1;          % time index
X = fft(x);         % DFT
k = 0:N-1;          % frequency index

disp("DFT X[k] = ");
disp(X);

% Plotting
figure;

subplot(3,1,1);
stem(n, x, 'filled');
title('Original Time-Domain Sequence');
xlabel('Time Index (n)');
ylabel('Amplitude');
grid on;
```

```
subplot(3,1,2);
stem(k, abs(X), 'filled');
title('Magnitude Response');
xlabel('Frequency Bin (k)');
ylabel('|X[k]|');
grid on;

subplot(3,1,3);
stem(k, angle(X), 'filled');
title('Phase Response');
xlabel('Frequency Bin (k)');
ylabel('Phase (radians)');
grid on;
```

```
DFT X[k] =
     3    -1    -1    -1
```

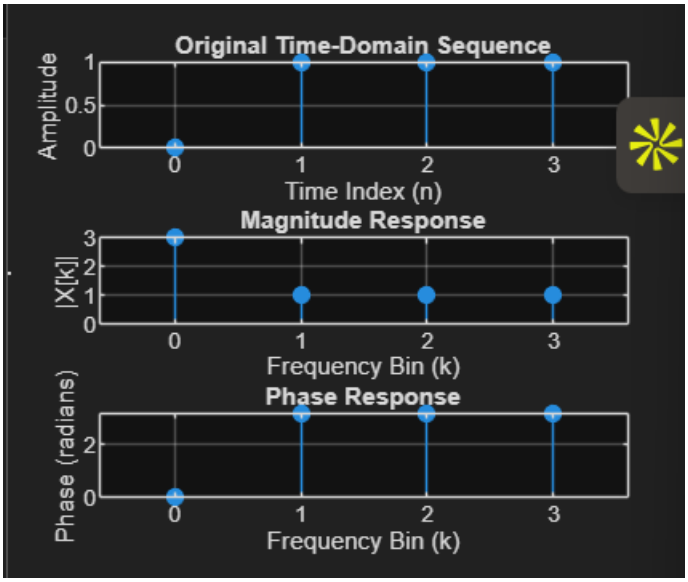
DTFT of [1 2 3 4]

```
clc;
```

```
clear;
close all;

x = [1 2 3 4];      % input sequence
N = length(x);

n = 0:N-1;          % time index
X = fft(x);         % DFT
k = 0:N-1;          % frequency index

disp("DFT X[k] = ");
disp(X);

% Plotting
figure;

subplot(3,1,1);
stem(n, x, 'filled');
title('Original Time-Domain Sequence');
```

```
xlabel('Time Index (n)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
stem(k, abs(X), 'filled');
title('Magnitude Response');
xlabel('Frequency Bin (k)');
ylabel('|X[k]|');
grid on;

subplot(3,1,3);
stem(k, angle(X), 'filled');
title('Phase Response');
xlabel('Frequency Bin (k)');
ylabel('Phase (radians)');
grid on;
```

```
DFT X[k] =
     3    -1    -1    -1
```



Filter wala question jo kal sir ne whatsapp par bheja tha

. Design a finite impulse response (FIR) low-pass filter with the following specifications: - Sampling frequency: 1000 Hz - Cutoff frequency: 200 Hz - Filter order : 31

```
fs = 1000;          % Sampling Frequency
fc = 200;           % Cutoff Frequency
N = 31;             % Filter Order

% Calculate normalized frequency (0 to 1, where 1 is fs/2)
Wn = fc / (fs/2);
```

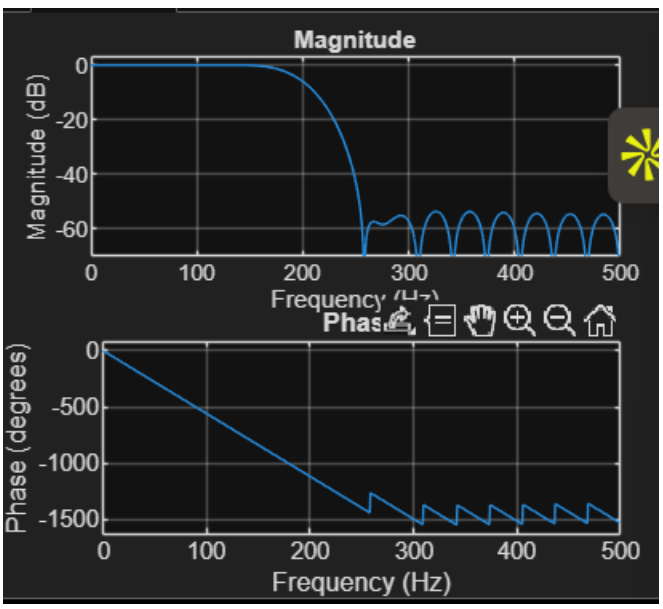
% Design the filter using a Hamming window (default)

b = fir1(N, Wn, 'low');

% Frequency response visualization

freqz(11111111111104
7+\2

b, 1, 1024, fs);



LOW pass filter (Hamming Window)

% FIR Low-Pass Filter Design (Hamming Window)

clc; clear; close all;

```
% 1. Filter Specifications
fs = 1000;          % Sampling frequency (Hz)
fc = 200;           % Cutoff frequency (Hz)
N = 31;             % Filter order
M = N + 1;          % Number of taps (coefficients)

% 2. Calculate Normalized Frequency
% Wn is the ratio of cutoff frequency to the Nyquist frequency (fs/2)
Wn = fc / (fs/2);

% 3. Determine Filter Coefficients (h[n])
% 'low' specifies a low-pass filter
% fir1 uses a Hamming window by default for the specified order N
h = fir1(N, Wn, 'low');

% Display the coefficients in the Command Window
fprintf('The %d filter coefficients h[n] are:\n', M);
```

```
disp(h');

% 4. Frequency Response Calculation
[H, f] = freqz(h, 1, 1024, fs);

% 5. Plotting Magnitude and Phase Response
figure;

% Magnitude Response (in dB)
subplot(2,1,1);
plot(f, 20*log10(abs(H)), 'LineWidth', 1.5);
grid on;
title(['Magnitude Response of Order ', num2str(N), ' FIR Low-
Pass Filter']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
axis([0 fs/2 -60 5]); % Focus on the passband and stopband

% Phase Response (in Degrees)
```

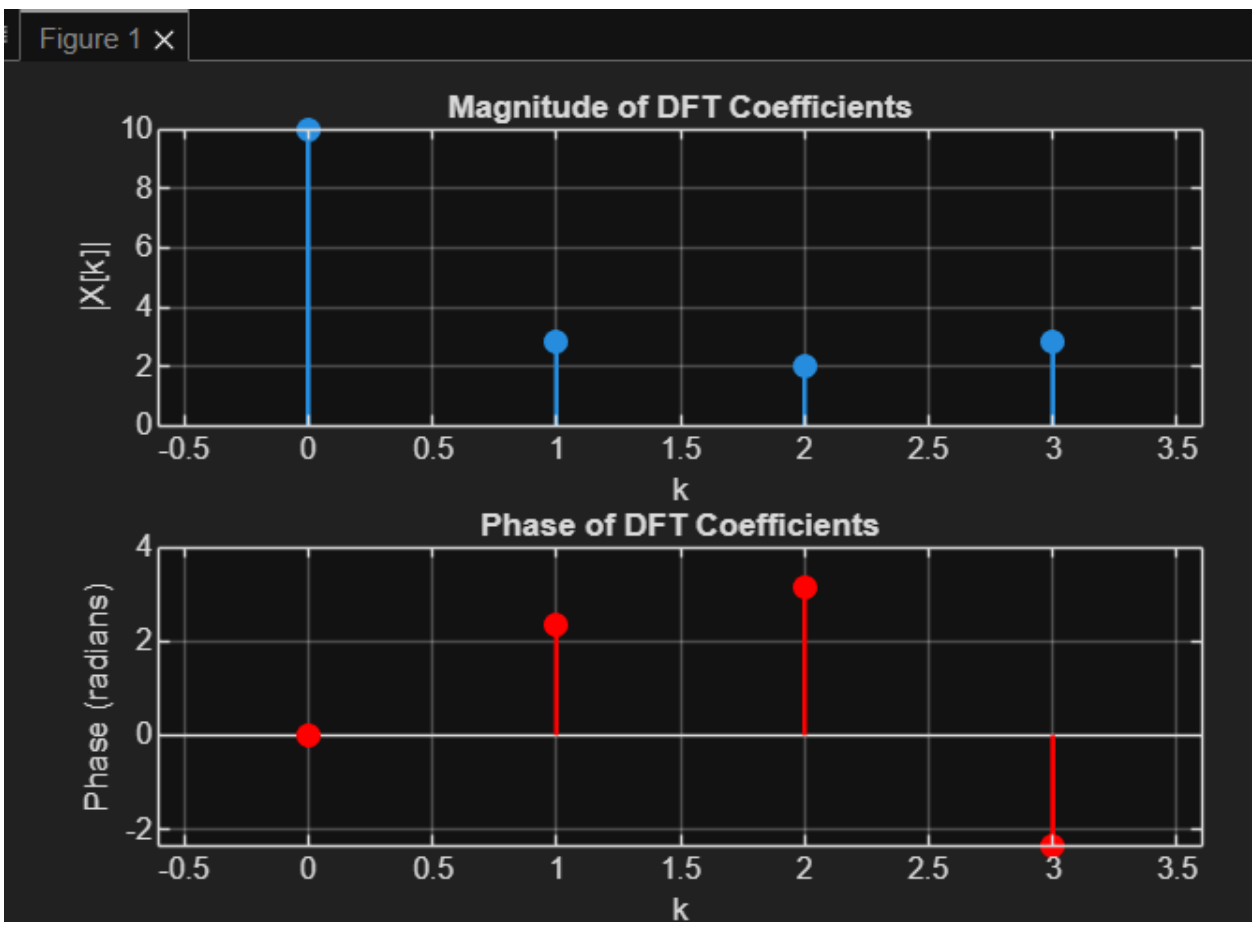
```
subplot(2,1,2);
plot(f, unwrap(angle(H)) * (180/pi), 'LineWidth', 1.5, 'Color', 'r');
grid on;
title('Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
```

```
The 32 filter coefficients h[n] are:
    0.0010
   -0.0012
   -0.0026
    0.0000
    0.0059
    0.0053
   -0.0075
   -0.0168
```

1. Calculate the 4-point DFT of the sequence x[n] = [1, 2, 3, 4] with MATLAB and Plot both the magnitude and phase of each DFT coefficient.

```
% Define the sequence
x = [1, 2, 3, 4];
N = length(x);

% Calculate the 4-point DFT
Xk = fft(x, N);

% Calculate Magnitude and Phase
magnitude = abs(Xk);
phase = angle(Xk); % Phase in radians

```

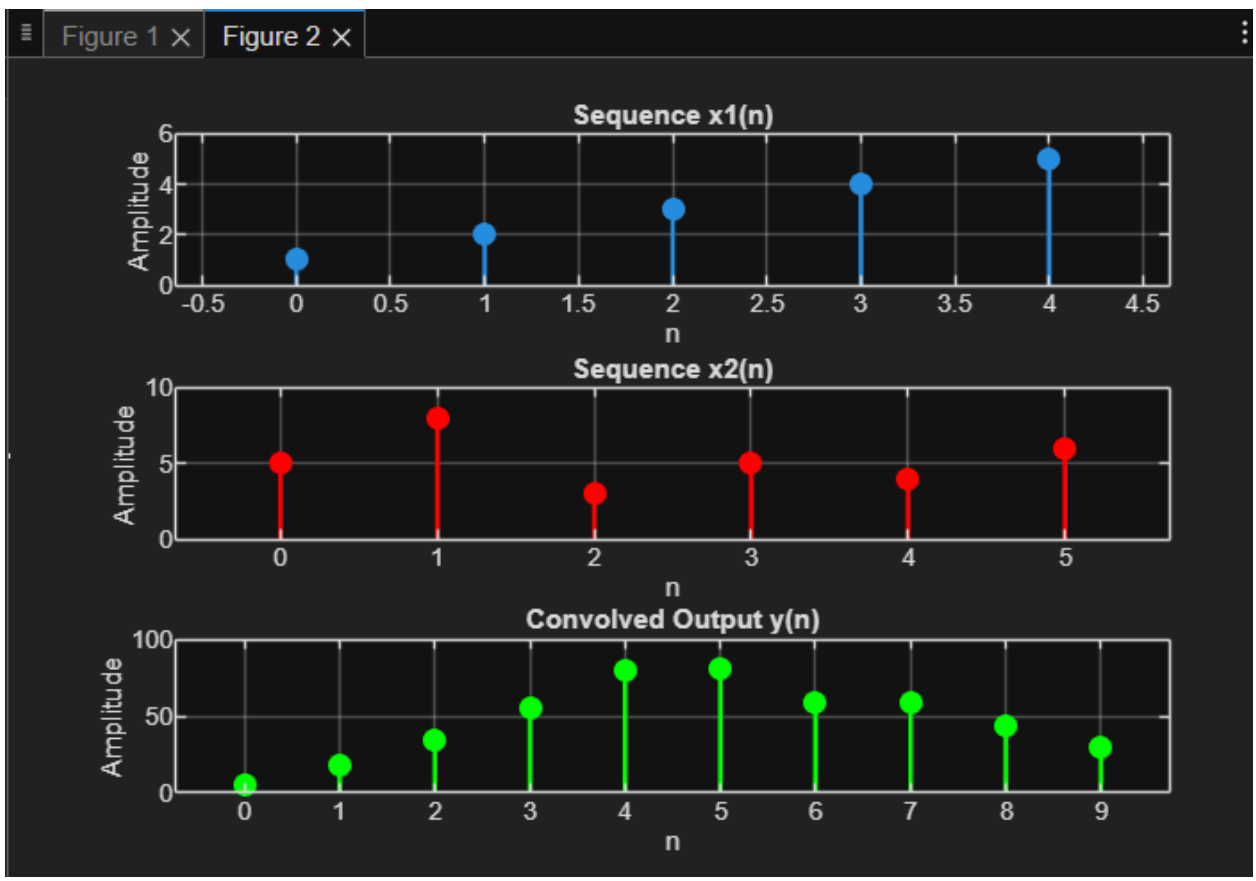
```
% Frequency axis for plotting
k = 0:N-1;

% Plotting
subplot(2,1,1);
stem(k, magnitude, 'filled', 'LineWidth', 1.5);
title('Magnitude of DFT Coefficients');
xlabel('k'); ylabel('|X[k]|');
grid on;

subplot(2,1,2);
stem(k, phase, 'filled', 'r', 'LineWidth', 1.5);
title('Phase of DFT Coefficients');
xlabel('k'); ylabel('Phase (radians)');
grid on;

% Display values in Command Window
disp('DFT Coefficients:');
disp(Xk);
```



2. Program to obtain Linear Convolution of two finite length sequences Enter the first sequence x1(n) = [1 2 3 4 5] Enter the second sequence x2(n) = [5 8 3 5 4 6] & plot the convolved output.

```
% Define the input sequences

x1 = [1, 2, 3, 4, 5];

x2 = [5, 8, 3, 5, 4, 6];


% Perform Linear Convolution

y = conv(x1, x2);


% Define time axes for plotting

n1 = 0:length(x1)-1;

n2 = 0:length(x2)-1;

ny = 0:length(y)-1;


% Plotting the sequences

figure;


subplot(3,1,1);

stem(n1, x1, 'filled', 'LineWidth', 1.5);

title('Sequence x1(n)');

xlabel('n'); ylabel('Amplitude');

grid on;


subplot(3,1,2);

stem(n2, x2, 'filled', 'r', 'LineWidth', 1.5);

title('Sequence x2(n)');
```

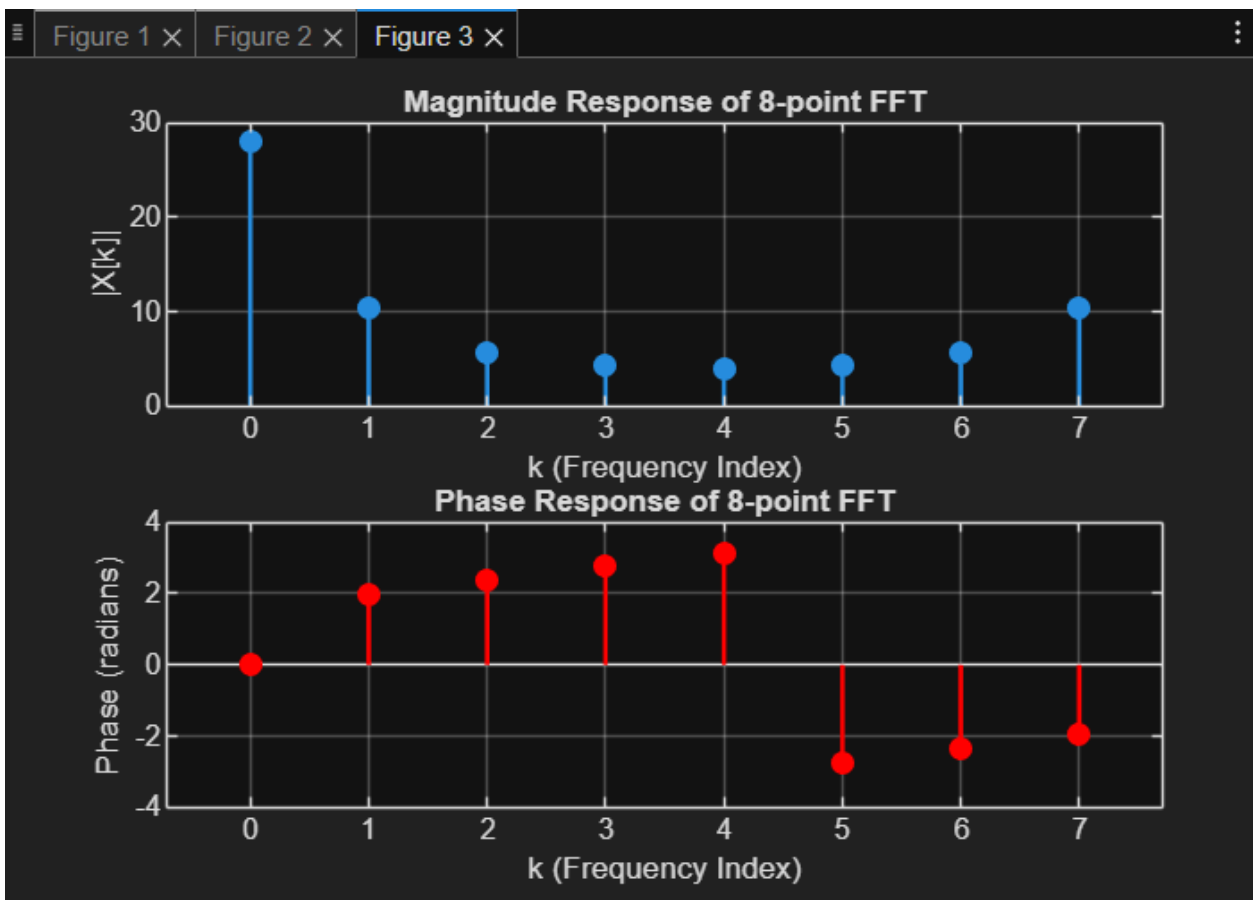
```
xlabel('n'); ylabel('Amplitude');
grid on;

subplot(3,1,3);
stem(ny, y, 'filled', 'g', 'LineWidth', 1.5);
title('Convolved Output y(n)');
xlabel('n'); ylabel('Amplitude');
grid on;

% Display result in command window
fprintf('The convolved sequence y(n) is:\n');
disp(y);
```



3. Given x(n) ={0, 1, 2, 3, 4, 5, 6, 7}, find X(k) using FFT with MATLAB and Plot both the magnitude and phase response.

```
% Define the input sequence
x = [0, 1, 2, 3, 4, 5, 6, 7];
N = length(x);

% Compute the 8-point FFT
Xk = fft(x);

% Calculate Magnitude and Phase
magnitude = abs(Xk);
phase = angle(Xk); % Result in radians

% Define the frequency index
k = 0:N-1;

% Plotting
figure;

% Magnitude Plot
subplot(2,1,1);
stem(k, magnitude, 'filled', 'LineWidth', 1.5);
title('Magnitude Response of 8-point FFT');
xlabel('k (Frequency Index)');
ylabel('|X[k]|');
grid on;
```

```
% Phase Plot
subplot(2,1,2);
stem(k, phase, 'filled', 'r', 'LineWidth', 1.5);
title('Phase Response of 8-point FFT');
xlabel('k (Frequency Index)');
ylabel('Phase (radians)');
grid on;

% Display the complex coefficients in the console
disp('FFT Coefficients X[k]:');
disp(Xk.');
```



19 - DSP
21 - Phalguna
23 - SDT
2 - RM
5 - [illegible]

③ ① $x[n] = [1, 2, 3, 4]$.

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi/N (nk)}.$$

$W_n = e^{-j\frac{2\pi}{N}}$ for 4 point DFT (N=4).

$$W_4 = e^{-j\frac{2\pi}{4}} = e^{-j\pi/2} = \cos\left(\frac{\pi}{2}\right) - j\sin\left(\frac{\pi}{2}\right) = -j$$

Twiddle factor

$W_4^0 = 1$

$W_4^1 = -j$

$W_4^2 = -1$

$W_4^3 = j$

$$X[k] = x[0]W_4^0 + x[1]W_4^{1k} + x[2]W_4^{2k} + x[3]W_4^{3k}.$$

$k=1$

$k=0$ $X[0] = 1(1) + 2(1) + 3(1) + 4(1) = 10$

$k=1$ $X[1] = 1(1) + 2(-j) + 3(-1) + 4(j) = -2+2j$

$k=2$ $X[2] = 1(1) + 2(-1) + 3(1) + 4(-1) = -2$

$k=3$ $X[3] = 1(1) + 2(j) + 3(-1) + 4(-j) = -2-2j$

$$\{10, -2+2j, -2, -2-2j\}$$

② $x_1(n) = [1\ 2\ 3\ 4\ 5]$ — 5

$x_2[n] = [5\ 8\ 3\ 5\ 4\ 6]$ — 6

$5+6-1$

$11-1 = 10$

	1	2	3	4	5
5	5	10	15	20	25.
8	8	16	24	32	40
3	3	6	9	12	15
5	5	10	15	20	25.
4	4	8	12	16	20
6	6	12	18	24	30.

$y(0) = 5$

$y(1) = 8+10 = 18$

$y(2) = 3+16+15 = 34$

$y(3) = 55$

$y(4) = 80$

$y(5) = 81$

$y(6) = 59$

$y(7) = 59$

$y(8) = 44$

$y(9) = 30$

$y(n) = [5, 18, 34, 55, 80, 81, 59, 59, 44, 30]$.

③. $x[n] = \{0, 1, 2, 3, 4, 5, 6, 7\}$.

$W_8^0 = 1$, $W_8^1 = \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$, $W_8^2 = -j$, $W_8^3 = \frac{-1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$, $W_8^4 = -1$

$W_8^5 = \frac{-1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$, $W_8^6 = j$, $W_8^7 = \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$

$$x(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}.$$

$$= \sum_{n=0}^{7} x(n) W_8^{kn}.$$

$$x(k) = \cancel{x(0) W_8^0} + x(1) W_8^k + x(2) W_8^{2k} + x(3) W_8^{3k} + x(4) W_8^{4k} + x(5) W_8^{5k} + x(6) W_8^{6k} + x(7) W_8^{7k}$$

$k=0$, $x(0) = 1 + 2 + 3 + 4 + 5 + 6 + 7 = 28.$

$k=1$ $x(1) = \frac{1}{\sqrt{2}}(1-j) + 2(-j)$

$= -4 + 9.657j$

$x(2) = -4 + 4j$

$x(3) = -4 - 4j$

$x[4] = -4$

$x[5]$ $-4 - 1.657j$

$x[6] = -4 - 4j$

$x[7] = -4 - 9.657j$